

Fig. 8: Screenshot of basic_twice.jpg

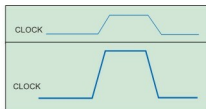


Fig. 9: Screenshot of basic_pixel.jpg

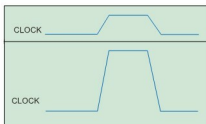


Fig. 10: Screenshot of basic_cell.jpg



Fig. 11: Screenshot of basic_line.jpg

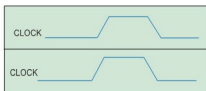


Fig. 12: Screenshot of font.jpg

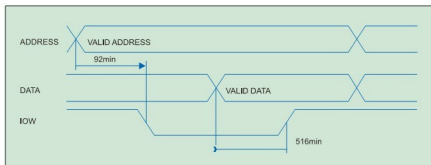


Fig. 13: Screenshot of delay.jpg

A semicolon can be used to separate a list of dependency signal changes for the same clock period, which is illustrated in the third clock period (refer Fig. 7) in the following example. The syntax is as given below:

```
CS=0, OE=0, LED=Z, DATA=Z.
CS=1, OE=1.
CS, OE => LED=ON;
DATA=DATA.
```

Command line options

Drawtimg has numerous command line options which can be seen by using the '-h' command at the terminal window. The supported output image formats are listed in ImageMagick website. ImageMagick is a free and open source software to view, edit and transform image files.

A PDF output of the basic.txt example can be created by specifying the .pdf filename extension in the terminal window as shown below:

```
$ drawtimg -o basic.pdf basic.txt
```

Here, the output will be the same as shown in Fig. 1, only file format is changed to .pdf.

The output image can be scaled with the '-x' option. To generate an image that is twice the default size, you can use the command as given below:

```
$ drawtimg -x 2 -o basic_twice.jpg
```

basic.txt

The output is as shown in Fig. 8

The upper clock separated by a thin line is the original clock signal shown here for clarity.

If you want to be explicit about the pixel width and height, the '-p' option followed by width 'x' height can be specified as given below:

```
$ drawtimg -p 250x75 -o basic_
pixel.jpg basic.txt
```

The output is as shown in Fig. 9

The default cell width and height are 64 and 48 pixels, respectively. The '-w' and '-c' options are used to modify them:

```
$ drawtimg -c 100 -o basic_cell.
jpg basic.txt
```

The output is as shown in Fig. 10

The line width can be changed using the '-l' option. For example:

```
$ drawtimg -l 3 -o basic_line.jpg
basic.txt
```

The output is as shown in Fig. 11

The default font size is 25 points. You can change it with the '-f' option. You can also specify a different font using the '-f' font option.

```
$ drawtimg -f 10 -o basic_font.jpg
basic.txt
```

The output is as shown in Fig. 12

Timing delays can be displayed in the output using '-tDS' construct between any two signals. In a simple I/O write cycle, the minimum time between a valid address and the I/O Write (IOW) strobe is shown as 92ns (minimum) as follows:

```
ADDRESS="VALID ADDRESS", DATA="",
IOW=1.
DATA="", IOW=0.
ADDRESS -92min> IOW;
DATA="VALID DATA".
IOW=1.
IOW -516min> DATA;
ADDRESS="", DATA="".
```

The output is as shown in Fig. 13

For a real-world example, let us consider the write cycle of a 68008 microprocessor. During the write cycle, the CPU first asserts the read/write (RW) signal for write operation, followed by the address and data signals. The data strobe (DS) signal is then asserted, and the CPU waits